

AUMANSH VIJAYENDRA GUPTA

AI/ML Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer skilled in **Deep Learning, Computer Vision, Federated Learning, and Secure Machine Learning**. Experienced in designing **end-to-end ML pipelines**, model training, evaluation, optimization, and deployment on cloud and edge systems. Built encrypted federated learning systems using **BFV/CKKS**, developed real-time **YOLO-based detection models**, and improved ML model accuracy by **80%**. Strong command of **TensorFlow, Keras, Scikit-learn, NumPy, Pandas, SQL**, and applied ML research.

TECHNICAL SKILLS

Programming: Python, SQL

Machine Learning: Supervised Learning, Unsupervised Learning, Feature Engineering, Model Optimization

Deep Learning: CNNs, DNNs, Data Augmentation, Transfer Learning, Object Detection (YOLOv8)

Computer Vision: OpenCV, Image Classification, Detection, Preprocessing, PCA, SVD

Federated Learning & Security: BFV/CKKS Encryption, Secure Aggregation, Privacy-Preserving ML

NLP: Text Processing, Tokenization, Chatbot Development (IBM Watsonx)

Tools & Platforms: TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Jupyter, Linux, Git, AWS, Power BI

Data Engineering: ETL Pipelines, Data Cleaning, AWS Cloud Processing

Soft Skills: Problem-solving, Agile, Documentation, Research Mindset

EDUCATION

Marwadi University, Rajkot, Gujarat

Bachelor of Technology in Computer Engineering (2022–2026)

CGPA: 8.26/10 — **7th Sem SGPA: 9.18**

Medium of Instruction: English

EXPERIENCE

Defence Electronics Applications Laboratory (DEAL), DRDO — Summer Trainee

May 2025 – July 2025 — Dehradun, India

- Developed a **computer vision-based Intruder Detection System** using PCA-SVD for feature extraction and KNN for classification.
- Implemented **real-time ML workflows**, optimizing inference time, preprocessing, and dimensionality reduction.
- Worked in a **high-security R&D environment** applying ML experimentation, testing, and model evaluation.

India Space Academy — Astronomy & Astrophysics Intern

June 2025 – July 2025 — Remote

- Conducted computational analysis on astrophysics datasets using **statistical modeling and regression**.
- Contributed to research involving **galaxy cluster mass estimation** and **Hubble constant prediction**.
- Applied Python-based data analysis workflows to improve scientific interpretation.

Afame Technologies — Machine Learning Engineer Intern

Apr 2024 – May 2024 — Bengaluru, India

- Improved predictive model accuracy by **80%** through feature engineering and hyperparameter tuning.
- Built and deployed ML models using **TensorFlow and Scikit-learn**.
- Followed Agile, CI/CD, and version control practices for model lifecycle management.

Edunet Foundation (IBM SkillsBuild) — AI/Cloud Intern

Jan 2024 – Feb 2024 — Rajkot, India

- Developed an **NLP chatbot** using IBM Watsonx Assistant with enhanced text preprocessing and API integration.
- Improved response accuracy by refining conversational flows and model logic.

PROJECTS

FLIP: Hybrid BFV-CKKS Federated Learning Framework

Python, TensorFlow, Scikit-learn, Homomorphic Encryption

- Implemented a **secure federated learning model** using BFV/CKKS encryption.
- Achieved **99.23% accuracy** on MNIST with encrypted model convergence.
- Designed distributed training loops, encrypted batching, and modular ML pipelines.

Naval Ship Detection (YOLOv8 + Federated Learning)

Python, Ultralytics, TensorFlow

- Built a **YOLOv8 real-time detection system** for naval vessel classification.
- Integrated federated learning for **privacy-preserving distributed training**.
- Reduced communication overhead while maintaining accuracy.

Data Waves — Data Engineering Capstone

AWS, SQL, Python

- Built end-to-end **data pipelines** for ingestion, transformation, validation, and querying.
- Reduced pipeline execution time by **30%** using SQL optimization.

Software Fault Tolerance with ML/DL

Python, Scikit-learn, NumPy

- Applied ML/DL algorithms for **software defect prediction**.
- Increased model accuracy through feature selection, cross-validation, and tuning.

CERTIFICATIONS

AWS Academy Graduate — Data Engineering

IBM Certified — Enterprise Data Science in Practice

Open Source Foundations

Prompt Engineering for Everyone

ACHIEVEMENTS

1st Rank — Infosys Python Coding Mock Assessment

Outstanding Performer — FOSSASIA “Open Source Meets Web3” Hackathon

89th Percentile Worldwide — ETS WheelBox Global Employability Test (IT Software Role)